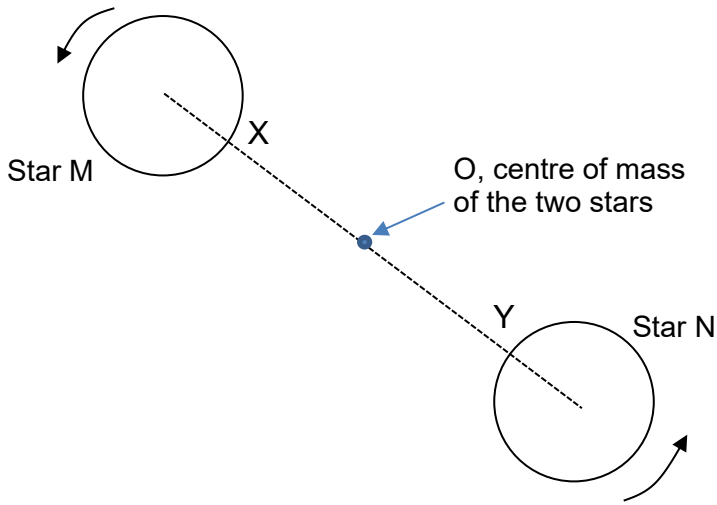
2 In a binary star system, two stars, each of equal mass 3.5×10^{30} kg, rotate about their common centre of mass O which is equidistant from the centres of the stars. The separation between the two centres of the stars is 2.0×10^{11} m.



(a) Define gravitational potential at a point.

.....

.....

..... [1]

(b) Calculate the gravitational potential at O, the centre of mass of the binary star system.

gravitational potential = ... J kg^{-1} [2]

(c) An asteroid passes through point O, at a speed v.

Determine the minimum speed of the asteroid if it is to escape from the gravitational pull of the binary star system.

minimum speed = …….. m s^{-1} [3]

(d) **(i)** On Fig. 2.1, sketch a graph showing the variation of gravitational potential along the line XY between the two stars. [2]

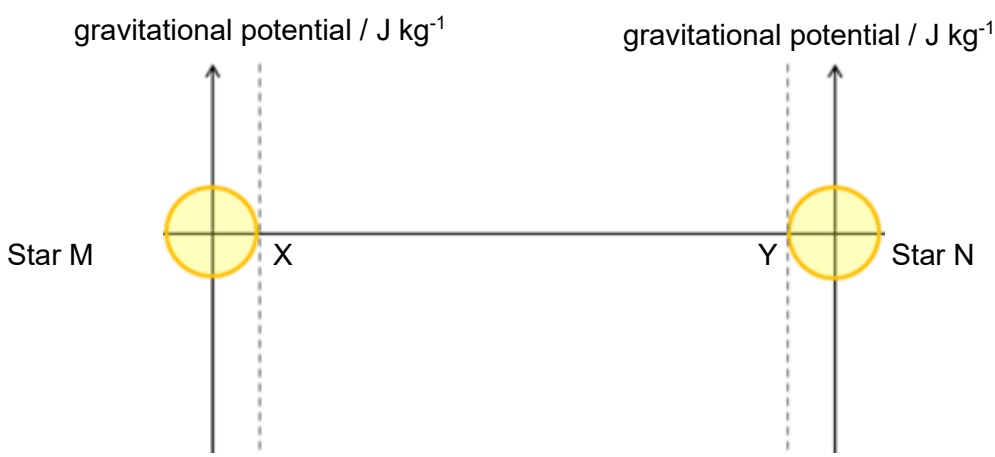


Fig. 2.1

(ii) Hence describe the variation in gravitational potential energy of an object moving from O towards star M.

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……

…… [2]